

GreenEye – QA Book

Early Test Cases – Sprint 2

1. Dummy Prediction API – Test Cases

Test ID	Test Case Description	Input	Expected Output	Priority
TC-API-01	Submit valid image as base64 payload	Base64-encoded JPG	200 OK, JSON with two prediction objects	High
TC-API-02	Submit unsupported image format	.gif or .bmp	400 Bad Request, error message	Medium
TC-API-03	Submit empty payload	Empty JSON {}	400 Bad Request, 'Missing image' error	High
TC-API-04	Submit corrupted base64 string	Malformed base64	422 Unprocessable Entity	Medium
TC-API-05	Simultaneous API calls (load test)	10+ concurrent requests	All return 200 OK within SLA	High
TC-API-06	One model fails internally	Simulate backend failure in Model B	Partial response + 'Model B unavailable' message	Medium
TC-API-07	Timeout scenario	Simulate high latency	Response in < 7s or timeout message	Medium

2. Frontend Upload Form – Test Cases

Test ID	Test Case Description	Input	Expected Output	Priority
TC-UI-01	Upload supported image file	Valid .jpg or .png under 5MB	Image preview shown within 1s	High
TC-UI-02	Upload unsupported file type	.pdf, .gif, .txt	File rejected, validation error displayed	High
TC-UI-03	Upload file >5MB	Large .jpg or .png	File rejected, error: 'File too large'	High
TC-UI-04	Upload image with too low resolution	50x50 px image	Warning: 'Low quality image may affect accuracy'	Medium
TC-UI-05	Remove uploaded image	Click 'Remove' button	Image preview cleared	High
TC-UI-06	Submit with no image uploaded	Click 'Submit'	Show error: 'Please upload an image before submitting'	High
TC-UI-07	Upload valid image, click Submit	Valid image → Submit	Button shows loading, result screen appears	High
TC-UI-08	Receive dual-model prediction result	Backend returns 2 results	Show results side-by-side with model labels	High
TC-UI-09	Upload invalid file, then valid file	Invalid → Valid	Error clears, new image accepted	Medium
TC-UI-10	Screen reader accessibility	Tab/ARIA navigation	All buttons/fields	Medium

			are focusable and labeled	
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3. API & Backend Endpoint – Test Cases

Test ID	Test Case Description	Input / Scenario	Expected Behavior	Priority
API-01	/predict endpoint returns 200 OK for valid input	Valid base64 image	Status 200 + valid JSON	High
API-02	Returns valid JSON with both model predictions	Valid input	Keys: species_model_A, confidence_model_A, etc.	High
API-03	Handles malformed or empty image payloads	Empty or {}	400 Bad Request	High
API-04	Handles corrupted base64	Corrupted string	422 Unprocessable Entity	Medium
API-05	Each prediction includes a species string and float confidence	Valid input	Species as string + confidence as float	High
API-06	Returns only one result if one model fails	Simulate Model B crash	Show Model A + failure message for B	High
API-07	Each model runs in parallel to reduce response time	Trigger and log timestamps	Total response < 7s	Medium
API-08	Logs each request with	Any request	Entry created in logs with	Medium

	timestamp and response time		metadata	
API-09	Blocks unsupported file types	Upload .exe, .pdf	400 error or rejection	High
API-10	Returns 500 only on unhandled internal failures	Trigger unexpected crash	500 + internal server error	High
API-11	Log species prediction frequencies for monitoring	Multiple valid calls	Species counts logged	Medium
API-12	Track prediction time per model	Any prediction call	Each model's duration recorded	Medium

4. Model Output Display – UI Test Cases

Test ID	Test Case Description	Input / Scenario	Expected Behavior	Priority
TC-UI-11	Display dual predictions	Backend returns 2 predictions	Species names and confidence scores shown side-by-side	High
TC-UI-12	Distinguish between models visually	Model A and Model B labels	Predictions clearly labeled (e.g., 'Fine-Tuned', 'Custom')	High
TC-UI-13	Show prediction with low confidence	Confidence = 0.40	Display subdued or warning icon near score	Medium
TC-UI-14	Handle conflicting predictions	Tomato vs. Potato	Display both results without error	Medium
TC-UI-15	Display only one prediction (Model B fails)	Model B fails	Show Model A result with fallback message	High
TC-UI-16	Show confidence score formatting	0.91235	Displayed as '91.2%'	High
TC-UI-17	Responsive layout	Resize screen	No overlap, clean mobile layout	Medium
TC-UI-18	Tooltip on hover for model info	Mouseover label	Tooltip: 'Model A: Fine-tuned ResNet'	Low
TC-UI-19	ARIA compliance for result display	Keyboard/tab access	Screen reader announces label and score	Medium
TC-UI-20	Spinner shown during prediction	Click 'Submit'	Loading spinner and disabled button appear	High

TC-UI-21	Retry button works after partial prediction failure	Model A returns, Model B fails	Retry button re-attempts Model B only	Medium
TC-UI-22	Prediction button disables during processing	Click submit	Submit button disabled until prediction complete	High
TC-UI-23	Spinner disappears after both results received or one fails gracefully	End of prediction	Spinner disappears, results shown	High

Traceability Matrix

1. Functional Requirements

Requirement ID	Description	Linked Test Cases
FR-1.1	Upload Single Image	TC-UI-01, TC-UI-07
FR-1.2	File Validation	TC-UI-02, TC-UI-03, TC-UI-04
FR-1.4	Show Preview	TC-UI-01, TC-UI-05
FR-1.5	Remove Uploaded Image	TC-UI-05
FR-2.1	Predict Plant Species	TC-API-01, TC-UI-08, TC-UI-11
FR-2.2	Display Confidence Score	TC-UI-08, TC-UI-16
FR-2.3	Handle Multi-leaf Fusion	Not implemented – dropped in scope update
FR-2.4	Two Models	TC-API-02, TC-UI-08, TC-UI-11
FR-2.5	Separate Predictions	TC-UI-11, TC-UI-12, TC-UI-14

FR-2.6	Tag Predictions	TC-UI-12, TC-UI-18
FR-3.1	Unsupported File Types	TC-UI-02, API-09
FR-3.2	Image Quality Warnings	TC-UI-04
FR-3.3	Prediction Failure Handling	TC-UI-15, TC-UI-21, API-06

2. Non-Functional Requirements

Requirement ID	Description	Linked Test Cases
NFR-1.1	Prediction response within 7 seconds	API-07, TC-UI-23
NFR-1.2	Support 50 simultaneous users	API-05
NFR-1.3	Models run in parallel	API-07
NFR-2.1	Completion without assistance	TC-UI-01, TC-UI-06, TC-UI-08
NFR-2.2	Accessibility compliance	TC-UI-10, TC-UI-19
NFR-2.3	Readable language	Manual verification only
NFR-2.4	Visual confirmation within 500ms	TC-UI-01, TC-UI-05
NFR-2.5	Tooltip/help text	TC-UI-18
NFR-2.6	Visual distinction of model predictions	TC-UI-12
NFR-2.7	Confidence normalization	TC-UI-16
NFR-3.1	95% uptime	Monitoring/log review – not unit tested
NFR-3.2	Retry failed predictions	TC-UI-21, API-06
NFR-3.3	Log prediction errors	API-08

NFR-3.4	Resilient image upload	TC-UI-04, SEC-02
NFR-3.5	97% accuracy under ideal conditions	Post-deployment test
NFR-3.6	One-model fallback	TC-UI-15, API-06
NFR-5.1	Cross-browser compatibility	Manual QA – not in TC list
NFR-5.2	Dockerized multi-cloud deployment	CI/CD deployment testing – out of scope for UI QA

Coverage Track

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Status	Meaning
 Covered	Fully linked to one or more test cases
 Partial	Only partially covered or pending test
 Missing	No test case currently linked

1. Functional Requirements

FR ID	Description	Coverage	Notes
FR-1.1	Upload Single Image	 Covered	Linked to UI test cases
FR-1.2	File Validation	 Covered	Size, type, resolution tested
FR-1.4	Show Preview	 Covered	Preview shown on upload
FR-1.5	Remove Uploaded Image	 Covered	Remove button tested

FR ID	Description	Coverage	Notes
FR-2.1	Predict Plant Species	✓ Covered	Dual-model output tested
FR-2.2	Display Confidence Score	✓ Covered	Confidence formatting tested
FR-2.3	Handle Multi-leaf Fusion	✗ Missing	Out of scope — dropped
FR-2.4	Two Models	✓ Covered	Two-model predictions and outputs tested
FR-2.5	Separate Predictions	✓ Covered	UI distinguishes results
FR-2.6	Tag Predictions	✓ Covered	Tooltip and labels tested
FR-3.1	Unsupported File Types	✓ Covered	Invalid formats handled
FR-3.2	Image Quality Warnings	✓ Covered	Low-res warning tested
FR-3.3	Prediction Failure Handling	✓ Covered	Partial prediction & retry tested

2. Non-functional Requirements

NFR ID	Description	Coverage	Notes
NFR-1.1	Prediction within 7s	✓ Covered	Timed test cases present
NFR-1.2	Support 50 users	⚠ Partial	Simulated, not stress tested
NFR-1.3	Parallel Model Execution	⚠ Partial	Timing logs assumed, no direct test
NFR-2.1	Completion without Help	✓ Covered	Usability test cases applied
NFR-2.2	Accessibility	✓ Covered	Screen reader and tab flow tested
NFR-2.3	Readability	⚠ Partial	Needs manual linguistic review
NFR-2.4	Fast Visual Feedback	✓ Covered	500ms limit tested
NFR-2.5	Tooltip Help	✓ Covered	Hover labels tested
NFR-2.6	Distinct Model Labels	✓ Covered	Dual labels verified

NFR ID	Description	Coverage	Notes
NFR-2.7	Confidence Normalization	✓ Covered	Formatting validated
NFR-3.1	95% Uptime	✗ Missing	Monitoring required
NFR-3.2	Retry Failed Predictions	✓ Covered	Retry logic tested
NFR-3.3	Log Errors	✓ Covered	Log presence confirmed
NFR-3.4	Upload Resilience	⚠ Partial	Manual re-upload allowed
NFR-3.5	97% Accuracy Benchmark	⚠ Partial	Prediction confidence percentage is provided
NFR-3.6	One-model fallback	✓ Covered	Partial response tested
NFR-5.1	Cross-browser Compatibility	⚠ Partial	Manual tests not recorded
NFR-5.2	Cloud Docker Deployment	✗ Missing	Outside QA scope